

CHAPTER-2

Tata CLiQ E-Commerce Database

2.0 INTRODUCTION

The entities required for the Tata CLiQ E-Commerce Database are identified based on the business requirements of an online shopping platform. The attributes of each entity, along with their Primary Keys, Foreign Keys, Unique Keys, and attribute constraints, are determined to ensure efficient data organization. The relationships among the entities are identified to represent customer interactions, product management, order processing, payment, and delivery activities within the system. The cardinality and participation constraints of each relationship are also analyzed to define how entities are associated with one another. This analysis serves as the foundation for designing the Entity Relationship (ER) Diagram, Relational Schema, and database implementation while ensuring data integrity, consistency, and minimal redundancy.

2.1 ENTITIES

Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	Not Null
Email	Unique
Phone	Unique
Password	Not Null
Address	Not Null

Entity: Seller

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	Not Null
Email	Unique
GST_Number	Unique

Entity: Brand

Attribute	Key Type

Brand_ID	Primary Key
Brand_Name	Unique

Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Brand_ID	Foreign Key
Category_ID	Foreign Key
Seller_ID	Foreign Key
Price	
Stock	

Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key

Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key

Entity: Shipment

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key

Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key

Entity: Wishlist

Attribute	Key Type
Wishlist_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key

2.2 ENTITY RELATIONSHIP

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	1:1	Partial/Total
Customer	Places	Order	1:M	Partial/Total
Seller	Sells	Product	1:M	Total/Total
Brand	Manufactures	Product	1:M	Partial/Total
Category	Categorizes	Product	1:M	Partial/Total
Order	Has	Payment	1:1	Total/Total

2.3 CONCLUSION

The entity analysis of the Tata CLiQ E-Commerce Database successfully identifies the major entities, their attributes, keys, and relationships required for an efficient online shopping system. The defined entities and their relationships ensure proper organization of customer, product, seller, order, payment, shipment, review, and wishlist information while maintaining data

integrity and minimizing redundancy. This analysis provides a strong foundation for designing the Entity Relationship (ER) Diagram, Relational Schema, and database implementation, enabling the development of a secure, reliable, and scalable e-commerce database system.